APEX PHARMACEUTICAL INGREDIENTS LTD.
QUALITY ASSURANCE DIVISION - DEVIATION & RCA SOURCE RECORD
Document Ref: QA-DEV-2026-0914 | Classification: Major Critical Deviation
Status: OPEN - Awaiting Root Cause Analysis & CAPA
Date of Deviation: October 25, 2026 | Plant: API Synthesis Unit 4, Reactor Bay 3

SECTION 1: PRODUCT & PROCESS OVERVIEW
Product Name: Atorvastatin Calcium Trihydrate API (Ph. Eur. / USP Grade)
Batch Number: ATC-2026-B089
Lot Number: LOT-2026-10-89
Manufacturing Stage: Final Stage Vacuum Cone Drying
Equipment ID: CVD-04 (Double Cone Rotary Vacuum Dryer, 2,000 L)
Batch Size: 450.0 kg | Target Yield: 94.5%

SECTION 2: DEVIATION DETAILS & OBSERVATION
On October 25, 2026, Quality Control completed testing of sample QC-2026-8941 representing batch ATC-2026-B089. The residual solvent analysis performed by Gas Chromatography (HS-GC-04) yielded an Out of Specification (OOS) result:
- Parameter: Residual Acetone
- Result Obtained: 5,840 ppm (Specification: <= 5,000 ppm per ICH Q3C Guideline)
- Residual Methanol: 1,820 ppm (Specification: <= 3,000 ppm - PASS)
- Related Substances Total: 0.28% (Specification: <= 0.50% - PASS)
- Assay (HPLC on anhydrous basis): 98.9% (Specification: 98.0% - 102.0% - PASS)

Immediate Containment:
Batch ATC-2026-B089 was quarantined in Warehouse Room ISO 8, Pallet ID #P-892 under QA yellow Hold Tape. No material from this batch has been dispatched or blended into finished lots.

SECTION 3: TIMELINE & OPERATIONAL ANOMALIES
- 2026-10-23 20:30 - Drying cycle initiated with setpoint jacket temperature 45°C and vacuum 15 mbar.
- 2026-10-24 01:15 - Facility main cooling tower pump tripped, causing condenser coolant flow to drop by 60%.
- 2026-10-24 01:30 to 05:00 - Vacuum in CVD-04 deteriorated from 18 mbar to 85 mbar. Concurrently, jacket temperature dipped to 38°C.
- 2026-10-24 06:00 - Maintenance restarted the coolant circulation; however, drying was discontinued at 08:30 based only on Loss on Drying (LOD) check (0.42%) without waiting for QC headspace solvent verification.
- Maintenance Inspection: Vacuum pump VP-02 showed worn mechanical seal faces and carbon residue.
- Condenser vent valve CV-04B was found partially fouled with micronized statin powder.

SECTION 4: EVIDENCE GAPS IDENTIFIED
1. Calibration record for jacket temperature sensor TT-102 has not been reviewed for past 6 months.
2. Preventive maintenance log for vacuum pump VP-02 seal replacement was overdue by 45 days.
3. SCADA electronic temperature and vacuum trend export between 01:00 and 06:00 was not attached to the batch record.
4. Gas Chromatography raw chromatogram integration parameters require second-analyst verification.